Royal Devon University Healthcare NHS Foundation Trust

Clinical Guideline for:

Investigation of Local Anaesthetic Allergy in Children and Young People

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SUMMARY

The aim of this guidance is:

- to enable the diagnosis of a local anaesthetic allergy or tolerance in a child or young person (0 to 18 years) suspected of reacting to a local anaesthetic agent.
- to maintain highest levels of safety at all times with least discomfort to the patient.

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1. INTRODUCTION AND BACKGROUND

2.1 The overall incidence of adverse effects during local anaesthesia is 2-3% (See Box 2 for differentials). True IgE-mediated allergic reactions to local anaesthetic (LA) agents (anaphylaxis, urticaria, bronchospasm) are extremely rare. (1-4) In a systematic review of LA challenges in the UK the incidence was found to be 0.97%.(5) A combination of skin prick testing (SPT), intradermal, and sub-cutaneous testing, using serial dilutions of anaesthetics, has long been used for the diagnosis of LA Allergy.

2.2 The predictive values of all other diagnostic tests (including specific IgE-testing) are either limited or unknown.

2.3 Provocation tests remain the gold standard diagnostic tool for the diagnosis of LA Allergy.

2. RATIONALE

Local anaesthetics, including topical creams, are commonly used and frequently required for dental and minor surgical procedures. To facilitate their safe administration in the future, a firm diagnosis of LA allergy is therefore required in patients who have experienced possible allergic symptoms to Las. This protocol is based on the protocols of other specialist allergy centres and our own clinical experience.

3. OVERVIEW

- Take a history (Box 1).
- Consider differentials (Box 2).
- Investigations – SPT PAH test and subcutaneous (s/c) injections (To organise through allergy nurses - can be done in clinic or Bramble Day Case Unit).

Fig.1: Local anaesthetic challenge procedure using Lidocaine (flowchart)

If testing a different LA, check with drug allergy lead consultant for dosage and strength.

- Thorough history (Box 1)
- Consider differentials (Box 2)

SPT to latex, chlorhexidine, relevant food or aeroallergens. **(May occur in OPD)**

If negative, proceed BUT if Latex history and Latex Allergy Tests are equivocal, proceed with L/A challenge and book a latex challenge on a later date.

SAFETY CHECKS (including emergency drugs) and CONSENT (Box 3)

If **Latex if POSITIVE** (and history compatible with Latex Allergy), **do not proceed** and treat as per Latex Allergy

Patch Test

1. **LMX4** – Apply a 2x small blobs of topical agent on each upper arm for 45 minutes (2 patches total)
2. **Chlorhexidine** soaked gauze for 15 minutes
3. SPT using local anaesthetic to be tested (ideally without adrenaline but not crucial)

Which LA? → See Box 4

If chlorhexidine patch is positive, inform consultant and update MyCare, but still proceed as per negative patch tests.

If topical patch/SPT is positive **(significant redness beyond area of patchs/ hives/swelling)**, inform consultant to decide which LA to test before proceeding.

The total subcutaneous adrenaline volume of this protocol is 1.6mls which approximates the 1.85mls recommended by Schatz et al (4)

POSITIVE OR UNEQUIVOCAL (See Box 5 and refer to "A guideline for the diagnosis and management of allergic reactions in children")
- Rebook to test different LA to find one tolerated. D/W consultant. Discharge planning (See Appendix 1).

Box 1: History prior to decision to challenge

involved at time of

Subcutaneous LA injections (Lidocaine)

Inject the LA where the topical patches were sited. One in each arm.
If lidocaine 2% with adrenaline is not available, use lidocaine 1% with adrenaline.

- Inject 0.1ml saline placebo (if nervous patient)
- Inject 0.1ml **Lidocaine 2% plus adrenaline *1/200 000***
- Wait 30min and take a full set of observations (HR, RR, O2 sats, BP) and examine injection site, chest and skin. Check with patient for subjective symptoms.
- Inject 1.5ml **Lidocaine 2% plus adrenaline 1/200 000**
- Inject one in each arm at the site of LMX4 patch.

?

b. Was this tolerated?

7. Does the history suggest possible;

ntibiotic allergy?

atex allergy?

c. Chlorhexidine allergy?

8. If there is no history of a previous LA reaction (i.e. patient presents because of a family history of LA reactions) or if the index LA reaction was not suggestive of immediate hypersensitivity, then the referring physician/dentist may make a clinical decision to allow for future administration of any LA (without further testing). This decision should only be made by a Clinician who is experienced in the field of Drug Allergy.

Observe for **1 hour** after final injection

NEGATIVE:
Discharge with clear management plan and letter (Appendix 1).

Box 2: Differential Diagnoses (6)	
Differential Diagnosis	**Potential Symptoms / Signs**
Vasovagal	Bradycardia, sweating, pallor. Transient in timing. Jerking movements if they are kept upright.
Arrhythmias	Sweating, pallor, collapse, abnormal pulse.
Seizures	Jerking movements.
Sympathetic stimulation	Tremor, tachycardia, hypertension hyperventilation, carpo-pedal spasm
Psychomotor	Atypical psychological behaviour, excessive anxiety
Operative trauma	Excessive local tissue damage, functional complement disorder
Adverse Reaction (non-allergic)	Drugs that may be associated with adverse symptoms e.g. adrenaline.
Drug Allergy	Antibiotics, Chlorhexidine, Povidine, Latex, Analgesics e.g. paracetamol, NSAID's, Opioids, Inhalation agents.
Allergy to latex, dental devices, acrylics and gels	Initially localised but may generalise
Box 3: Pre-challenge safety checklist	

Pre challenge check list:

- Airway equipment available and in working order.
- Patient weight measured.
- Emergency drugs available and prescribed.
- Baseline observations - TPR, BP, Sa02, PEFR (if the child is > 5 years) together with auscultation of the chest should be recorded.
- Is the child fit for a challenge today (free from coughs, colds, allergic rhinitis, asthma exacerbation and afebrile)? If not, postpone / re-book.
- Consent (in line with trust guidelines).
- Drug challenge drugs prescribed and checked by nurse / doctor administering.
- Have a copy of "**A guideline for the diagnosis and management of allergic reactions in children**" easily accessible.

- **Any concerns:** Discuss with the consultant covering challenges or Paediatric allergy consultant. An experienced StR in Paediatric Allergy or Allergy Clinical Nurse Specialist may also be able to help.
- Latex allergy - avoid all latex products during admission.
- The challenge should be discontinued at any stage where a reaction occurs (see Box 5 below for signs of an allergic reaction) and treated appropriately as per the allergic reaction guidelines.

Emergency drugs – refer to BNFc

Drug	Dose
Chlorphenamine (IV, S/C, IM)	12 - 17 years 10mg 6 – 11 years 5mg 6 months to 5 years 2.5mg 1 – 5 months 250mcg / kg
Nebulised salbutamol	2.5-5mg
Hydrocortisone	12 - 17 years 200mg 6 – 11 years 100mg 6 months to 5 years 50mg 1 - 5 months 25mg
Prednisolone	2mg/kg po (max 40mg)
IM Adrenaline 1 in 1000 (Adrenaline auto-injectors available or draw up from syringe)	>12 years 500mcg (0.5ml) 6-12 years 300mcg (0.3ml) <6 years 150mcg (0.15ml)
Nebulised Adrenaline 1 in 1000 (1mg/ml)	1ml diluted with 3ml sodium chloride 0.9%
Fluid boluses	20ml/kg 0.9% sodium chloride

Age	Cetirizine dose (po)
Under 2 years	0.25mg/kg bd
2 – 6 years	2.5mg bd or 5 mg od
Over 6 years	5mg bd or 10mg od

Box 4: Which LA to choose?	
Implicated LA	**Challenge to:**
• **Unknown**	• **2% Lidocaine plus adrenaline** • **1/200 000 mix.**
• **Amide**(Non anaphylactic) Bupivacaine Etidocaine Levobupivacaine Lidocaine Mepivacaine Prilocaine Ropivacaine	• Test to **same Amide LA UNLESS** the reaction was unequivocally that of Anaphylaxis (i.e. severe, or life threatening allergic reaction). • EMLA® (Prilocaine 2.5% / Lidocaine 2.5% mix) – Can request from pharmacy if required.
• **Amide** (**Anaphylaxis)** Bupivacaine Etidocaine Levobupivacaine Lidocaine Mepivacaine Prilocaine Ropivacaine	• Test to a **different** (but commonly used) **Amide LA** as cross reactivity is rare within Amide LA's (always to be planned in conjunction with the Allergy Consultant). If risk deemed too high then perform challenge with an **Ester LA.**
• **Ester** Group: Benzocaine Chloroprocaine Cocaine Procaine Tetracaine (Ametop®)	• Testing will still be performed to **2% Lidocaine plus adrenaline** **1/200 000 mix.** • Ametop® (tetracaine) – Can request from pharmacy if required.

- Patch testing data suggests there is cross-reactivity among the esters but not the amides.
- If lidocaine 2% with adrenaline is not available, use lidocaine 1% with adrenaline.

Box 5: Signs of allergic reaction LEAP study, 2006 and Sampson HA. Anaphylaxis and emergency treatment, 2003

	Major criteria, i.e. reactions requiring IM adrenaline	**Minor reactions**
Respiratory signs (at least one of the following)	• Respiratory arrest • Throat itching or tightness • Hoarseness • Barky cough • Difficulty swallowing • Inability to speak • Stridor • Dyspnoea • Apnoea	• Nasal congestion • Sneezing • Rhinorrhea lasting for ≥3 mins • Rubbing of nose or eyes for ≥3 mins
Cardiovascular signs	• Cardiac arrest • Severe Bradycardia or hypotension, not associated with a vasovagal episode • Mild dysrhythmia and/or hypotension not associated with a vasovagal episode	• Tachycardia (increase of >15 beats/min)
Neurological signs	• Loss of consciousness • Feeling of impending doom	• Change in activity level • Anxiety • Light headedness
Skin signs		• Urticarial, pruritic rash • Angioedema • Persistent scratching lasting for ≥3 mins • Localised or generalised itching, flushing, urticaria
GI signs		• Oral itching or tingling • Mild lip swelling • Nausea • Vomiting • Diarrhoea

Clinical manifestations suggestive of true IgE-mediated reactivity would include urticaria, bronchospasm or distant angioedema occurring within 1 hour of LA injection.

5. REFERENCES

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APPENDIX 1: LOCAL ANAESTHETIC CHALLENGE RECORD SHEET on MyCare

Local Anaesthetic Challenge Record Sheet

Date: DD/MM/YYYY

Doctor requesting challenge: Weight:

Consultant supervising: Adrenaline dose:

Nurse Observing Child:

Antihistamine stopped prior to challenge: Yes / No (Cetirizine 1 wk before, Piriton 48 hours before)

CHECK LIST	Please tick √
Oxygen and Suction checked	
Anaphylaxis drugs checked and available	
Anaphylaxis protocol readily available	
No intercurrent illness	
No exacerbations of asthma in last week	

ASSESSMENT PRE CHALLENGE

History of previous reaction / anaphylaxis:

Asthma: (Regular inhalers, last exacerbation requiring steroids, Triggers)

Inhalers:

Suspected Local anaesthetic agent:

Adrenaline auto injector? Yes / No Training? Yes / No

Patient name: .....
NHS no: .....
Hospital no: ..... Please affix patient ID label within this box
DOB: .....

FITNESS FOR CHALLENGE / EXAMINATION	
Chest: Skin Condition: Any other Allergies:	
SPT Allergen on Challenge Day	**Result**
Latex	
Chlorhexidine	
Lignocaine	
Relevant other local anaesthetic	
Relevant foods	
Relevant aeroallergens	
PATCH TESTING on Challenge Day	**Result**
LMX4	
Chlorhexidine	
EMLA	
Sucutaneous Injections	**Result (if positive, note size increase)**
Saline placebo 0.1ml	
Lidocaine 2% plus Adrenaline 1/200000 0.1ml	
Lidocaine 2% plus Adrenaline 1/200000 1.5ml	

Patient name: .....
NHS no: .....
Hospital no: ..... Please affix patient ID label within this box
DOB: .....

PAEDIATRIC ALLERGY SERVICE
CHALLENGE OBSERVATION RECORD – Challenge:
Inclusion of age appropriate PEWS score sheet

Time	***Dose***	***Comments***
HH:MM		
HH:MM		
HH:MM		

APPENDIX 2: DISCHARGE PLANNING

1. Update **all** allergies on electronic records.
2. Write a discharge letter including the following points:
    a. The patch testing performed and results.
    b. The skin prick testing performed and results.
    c. The subcutaneous LA injections performed and results.
    d. Whether the challenge was considered negative or positive and to which drugs.
    e. State which LA drugs can be safely administered in the future.
    f. If the challenge was positive, state that until we challenge to an alternative LA, local anaesthetics should be avoided.
    g. If history was of a reaction to an ester and we challenged to lidocaine with no reaction, state that esters should be avoided but lidocaine is safe to use.
    h. Highlight the telephone contact numbers on the Paediatric Allergy Department letterhead for contacting the department regarding any queries.
3. Copy the discharge letter to the parents (to show health care professionals they encounter in different trusts), the GP, the referring doctor / dentist and have a copy in the patient records.
4. Book a date for repeat challenge to an alternate LA if the challenge was positive.
5. Book a latex challenge if there is a strongly suggestive history, despite negative skin prick test.

APPENDIX 3: EXAMPLE LETTER

Dear

Thank you for referring child's name for a local anaesthetic challenge. On date we carried out the following investigations:

- Patch testing to ++ and chlorhexidine which were *positive / negative*. *(delete appropriately)*
- Skin prick testing to local anaesthetic which was *positive / negative. (delete appropriately)*
- Subcutaneous injection of local anaesthetic. 1.6ml in total over two injections of 0.1ml and 1.5ml. This challenge was *positive with (insert symptoms / reaction) / negative (delete appropriately).*
- *Negative Challenge (delete appropriately)*

We are pleased to inform you child's name did not react at any point to any of the above drugs.

We consider this to be a negative challenge and child's name does not appear to be allergic to the local anaesthetic challenged. Local anaesthetic may therefore safely be administered in the future.

Were you to administer any local anaesthetics in the future, we would appreciate feedback with regards to child's name's progress.

Please do not hesitate to contact us on the above telephone numbers, Monday to Friday (0900 – 1700), if you have any queries.

Yours sincerely…

DELETE ABOVE OR BELOW Negative Allergy Letter above, Positive Allergy Letter below

- *Positive Challenge (delete appropriately)*

Unfortunately, child's name reacted to *patch testing / skin prick testing / subcutaneous injection* of local anaesthetic as described above. This local anaesthetic should be strictly avoided in the future. A further challenge appointment has been booked on date to test for a different local anaesthetic that might be tolerated. The patient was treated with *insert allergic reaction management provided.*

Until a safe local anaesthetic has been established, please avoid local anaesthetic medications.

Please do not hesitate to contact us on the above telephone numbers, Monday to Friday (0900 – 1700), if you have any queries.

Yours sincerely…